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Next Steps in Estimating Facility Renewal Costs: Proceedings of a Workshop—in Brief (2026)

DETAILS

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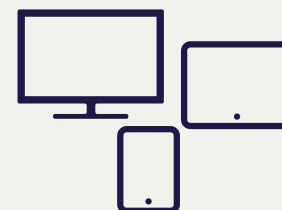
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Next Steps in Estimating Facility Renewal Costs

INTRODUCTION

The National Academies of Sciences, Engineering, and Medicine convened a two-day workshop, Next Steps in Estimating Facility Renewal Costs, to examine methods and data resources for improving cost estimates of facility renewal and modernization requirements. Organized by an ad hoc planning committee under the Board on Infrastructure and the Constructed Environment, the event brought together engineers, economists, and facility managers from federal agencies, national laboratories, academia, and industry. Over the course of the workshop, participants discussed approaches to depreciation modeling, estimation of service life, verification and validation (V&V) of analytical models, and the integration of large-scale condition assessment data into decision-support systems. They also explored challenges related to data quality and the use of geometric depreciation and other modeling approaches to forecast renewal requirements more effectively. Cameron Oskvig, National Academies, explained the workshop's aims to explore technical advances in modeling and data resources for facility renewal cost estimation and to identify promising directions for further research and development (R&D).

The discussions built on the 2023 National Academies report *Strategies to Renew Federal Facilities*, which identified the need for more rigorous approaches to estimating renewal requirements. Recommendation 4 of that report

(see Box 1) called for developing and applying improved analytic models—such as depreciation-based and recapitalization approaches—to forecast restoration and modernization costs more accurately. Jack Dempsey, Lindahl Reed, who served on the committee that authored the report, explained that although federal agencies have long modeled sustainment and recapitalization needs, they still lack a consistent, empirically grounded framework for predicting renewal costs. The workshop was convened to examine how new data resources and modeling methods could address that gap.

Kim Budil, Lawrence Livermore National Laboratory (LLNL), opened the workshop by explaining that the laboratory sponsored the event to advance the development of data-driven, verifiable models for estimating facility renewal requirements. She added that the National Nuclear Security Administration's (NNSA's) stewardship mission depends on aging facilities originally built for Cold War programs, making more reliable forecasting methods essential for investment planning across the enterprise. Budil also observed that aging infrastructure, evolving technologies, and constrained resources make the challenge of maintaining and renewing critical assets even more complex.

This Proceedings of a Workshop—in Brief summarizes presentations and discussions. It does not present findings or

BOX 1**RECOMMENDATION 4: IMPROVE FEDERAL FACILITY MODELS, DATA, AND MEASURES FROM NATIONAL ACADEMIES REPORT: STRATEGIES TO RENEW FEDERAL FACILITIES**

The Office of Management and Budget (OMB) should clarify guidance requiring agency senior real property officials to improve cost estimates of renewal requirements. Currently, there is no broadly accepted approach to estimating renewal costs, which diminishes the credibility of renewal decision making. After considering two of the methods available, the committee recommends the following:

- Senior real property officials should adopt an economic depreciation¹ approach for estimating renewal costs, tailorable to each agency's facility portfolio. As a starting point, the model could be simplified to a set of cost factors by facility type, analogous to the Department of Defense Facility Sustainment Model.
- Agencies should include existing dated depreciation rates and service lives in the economic depreciation approach review by using a schedule established for the revision of depreciation rate and service life data used in depreciation models, which is currently provided by the Department of Commerce's Bureau of Economic Analysis.

Furthermore, the General Services Administration (GSA), in coordination with the Federal Real Property Council and under the direction of OMB, should create an independent database of component inventories for federal facilities, beginning with the extensive data collected for the Builder system, and make it available to qualified users and accessible by popular capital planning and facility management systems. The senior real property officials of all agencies would submit information to GSA for compiling, as directed by executive requirement.

¹Economic depreciation refers to how an asset (structures, for example) declines in productivity over time. It is contrasted with tax depreciation, which is whatever the tax authorities allow you to use when filing income taxes. SOURCE: National Academies of Sciences, Engineering, and Medicine. 2023. *Strategies to Renew Federal Facilities*. Washington, DC: National Academies Press. <https://doi.org/10.17226/26806>.

recommendations of the National Academies or the workshop planning committee. This summary is organized by the overarching themes of the event rather than in the order of presentations.

STRATEGIC AND MISSION VALUE OF FACILITIES

Presenters and discussants framed facilities as mission-enabling assets whose condition, reliability, and suitability can shape readiness, operational continuity, and mission risk. Over the course of the workshop, participants returned to the same practical challenge: renewal models are most useful when they connect facility needs and investment

choices to mission consequences rather than treating facilities as a separate maintenance problem.

Michael McAndrew, former senior official, Department of Defense (DoD), observed that specialized federal infrastructure—such as laboratories, shipyards, and testing facilities—requires more than routine maintenance planning. He noted that many of these complex assets have long exceeded the service lives they were originally designed for, leading agencies to rely on incremental repairs rather than reinvestment. McAndrew explained that facility sustainment models typically succeed at estimating maintenance and recapitalization needs but do not necessarily connect those

needs to mission outcomes. Within most agencies, he said, mission activities drive most budget decisions, leaving less funding available for infrastructure. As a result, even sophisticated sustainment models have limited influence on priorities. McAndrew illustrated this disconnect with an example of flooding at the U.S. Military Academy at West Point, which he said was blamed on climate even though it resulted from underfunded routine maintenance. McAndrew used the example to show how infrastructure problems may only receive attention after a failure occurs and then be mischaracterized.

James McConnell, NNSA, emphasized that predictive modeling can improve both efficiency and mission alignment, as well as avoid more costly and reactive repairs. Expanding predictive modeling of condition assessments, he said, can help identify high-risk systems before they fail and allocate limited repair funding where it will have the greatest mission impact. McConnell added that the Mission Dependency Index (MDI) used within NNSA provides a mechanism to weight facilities by their operational importance, allowing predictive analytics to focus on assets whose degradation would most directly threaten mission continuity. Models need to be linked demonstrably to successful project execution because decision makers lose confidence in systems not grounded in the real world.

Workshop planning committee member and session moderator John Conger, Conger Strategies & Solutions, observed that the value of any renewal model lies in how clearly it links funding levels to operational consequences. He noted that leaders often see deferred maintenance figures or condition indices but rarely the evidence connecting those numbers to mission outcomes; he emphasized that models must help demonstrate that relationship. Drawing on DoD experience, Conger explained that facilities can degrade for years without visible effect until a catastrophic failure suddenly disrupts operations, and he noted that such crises often drive investment decisions. He encouraged model developers to capture both the probability and the consequences of failure so that resource decisions reflect measurable mission risk rather than abstract backlog totals. Conger later remarked that documenting real facility failures, rather than relying solely on indices, can help demonstrate how underinvestment translates into mission risk.

Workshop planning committee member Craig Crutchfield, former Office of Management and Budget (OMB), emphasized the importance of clear definitions when using mission-based indices, noting that agencies may otherwise classify too many assets as mission critical. He added that the larger challenge is showing how facility investments reduce risk to the overall agency mission.

MEASURING FUNCTIONAL OBSOLESCENCE AND MISSION RISK

Speakers emphasized that renewal modeling must capture not only the physical degradation of assets but also the loss of functionality over time. Several workshop participants described situations in which facilities remained structurally sound yet no longer supported the technologies, safety standards, or workflows required for their missions.

Workshop planning committee member Eric Dillinger, Woolpert, who moderated the session in which participants discussed what constitutes renewal, explained that renewal is often viewed too narrowly as condition improvement when it should also encompass modernization and mission-driven changes. He cautioned that this narrow view overlooks “mission obsolescence,” which occurs when a building or system can no longer support its intended function even if it remains in good repair. During the discussion, workshop participants illustrated this problem using examples from federal laboratories and defense installations where evolving modernization demands—such as new power requirements, seismic codes, or security standards—forced agencies to decide whether to retrofit existing facilities or construct new ones.

Dillinger noted that measurement systems designed for condition assessment are not necessarily sensitive to functional limitations. He remarked that existing metrics may identify defects but miss situations in which a system operates flawlessly at a task that no longer meets mission needs. He observed that assessments focused solely on physical condition can overlook if facility systems still meet the functional demands of evolving missions, such as changes in process requirements or operating environments.

Lucia Garsys, Siela Strategies and former Hillsborough County, Florida, noted that local governments face similar renewal challenges in managing large, aging infrastructure

networks. She described how the county's extensive road system—stretching the equivalent distance from Tampa to San Francisco and back—illustrates the consequences of delayed maintenance. Using pavement management as an example, she explained that postponing resurfacing allows deterioration to reach the point at which entire roadways must be torn up and rebuilt, multiplying costs and service disruptions. Her example showed how functionality and service quality can erode long before structural failure, reinforcing the need for timely renewal to sustain mission performance.

Gary Walker, Seattle Children's Hospital, described how the health care sector has adopted risk-based frameworks that explicitly connect facility performance to operational consequences. He outlined three factors that determine when renewal is preferred to new construction: (1) location relative to patient demographics, (2) compliance requirements such as seismic performance, and (3) physical constraints like floor-to-floor height that limit system upgrades. Hospitals, he said, routinely evaluate how space constraints or outdated mechanical systems could disrupt patient care, and they use redundancy scores to measure how many component failures a system can sustain before patient care is compromised. His example illustrated how mission criticality assessments can clarify the operational risks associated with facility underperformance.

Ellen Crews, Construction Specifications Institute, added that renewal decisions must also consider the quality of the documentation for existing assets. Incomplete construction records documenting the facility as built, she said, can mask hidden deficiencies or overstate a facility's capability to accommodate new systems. She explained that missing records can lead teams to assume unused capacity that in fact does not exist. In other sessions, workshop participants noted that gaps in facility data are sometimes filled with default values—such as “100,” implying perfect condition—when the actual information is missing, which further distorts renewal analyses. Crews emphasized that more specific requirements for turning over facility information at closeout would reduce uncertainty in both condition and functionality assessments.

Building on these points, workshop participants discussed how to represent mission risk quantitatively within renewal models. McConnell described the MDI, which assigns scores

to facilities based on how loss of function would affect mission performance. Workshop participants noted that such indices could help agencies prioritize sustainment and renewal investments according to mission impact.

One workshop participant suggested giving leaders a “risk lever” that allows them to set acceptable risk thresholds rather than confronting them with a single aggregate backlog figure. Related discussion emphasized that model results become more useful when they are tied to mission risk and consequences, rather than presented as maintenance backlog totals alone.

Dillinger emphasized the need to connect facility data to organizational mission if renewal analysis is to influence decisions. Workshop planning committee member Michael Grussing, U.S. Army Engineer Research and Development Center (ERDC), added that building this connection will require collaboration among engineers, mission operators, and other stakeholders.

Speakers concluded that measuring functional obsolescence and mission risk represents the next frontier for facility renewal modeling. By aligning technical metrics with mission performance, agencies can better focus limited resources on the assets whose failure would have the greatest impact. To support this prioritization, Dillinger added that measuring functional adequacy requires a standard or target to measure against, which makes it possible to assess the difference between the current and target performances.

QUANTIFYING FACILITY SERVICE LIFE

Service life was repeatedly described by workshop participants as one of the most influential parameters in renewal modeling. Speakers noted that service life underpins both economic depreciation approaches and condition-based projections, and that model outputs can be highly sensitive to the service-life values used.

Dylan Schmerer, LLNL, presented experimental data from BUILDER Sustainment Management System (SMS), a facility condition-assessment and forecasting tool, illustrating how strongly equipment renewal cost forecasts depend on assumed service life. Extending the service-life parameter by five years for all pieces of equipment resulted in a 14 percent reduction in cumulative deferred maintenance by the end of the 10-year

projection, illustrating the sensitivity of deferred maintenance to service-life assumptions. Manufacturer-provided service lives are typically shorter than those observed in the field; LLNL often observes replacement ages that are 30 percent longer than those specified by manufacturers. In response, Schmerer recommended limiting deferred maintenance for mission-critical facilities to roughly 5 percent of replacement value and prioritizing high-consequence equipment within that limit. He explained that these findings underscore the need to ground all renewal models in empirically verified service-life distributions rather than manufacturer warranties or design-life assumptions.

Louis “Buddy” Bartels, U.S. Army ERDC, noted that the U.S. Army’s SMSs have already accumulated millions of component-level observations that could serve this purpose. BUILDER SMS and related systems, he said, record the condition and age of facility components at the time of inspection, enabling analysts to manage sustainment requirements. Grussing added that data from SMSs and Facility Investment Analytics (FIA) can be integrated with other analytic tools to support comprehensive assessments of facility renewal and long-term planning.

Workshop planning committee member Peter Lufkin, Pomar Lane, underscored that service-life estimation is the linchpin of geometric depreciation models and emphasized that differences among renewal models are secondary to the assumptions that drive them. He stated that renewal modeling outcomes are “driven by service life” and that service-life assumptions determine results regardless of whether geometric depreciation or other approaches are used. He further noted that many depreciation rates in current use are based on studies from the 1970s that remain in circulation, meaning that economic forecasts continue to rely on decades-old data derived from commercial experience rather than contemporary federal portfolios.

Grussing said existing conditions and investment data could be used to assess and compare service-life assumptions. He noted that data from SMSs and FIA can be integrated with other analytic tools to support comprehensive assessments of facility renewal and long-term planning.

Workshop planning committee member Anand Desai, Clarivate, added that uncertainty in service life should be

treated explicitly rather than hidden within point estimates. He said forecasts could present a range of results to show how differing assumptions affect outcomes, giving decision makers a clearer sense of potential variance in long-term investment needs. He added that displaying those ranges helps decision makers understand risk and variance in long-term planning.

Speakers observed that using comparable service-life and depreciation assumptions across agencies would improve consistency in modeling results and allow parameters to be refined as new performance data accumulate. Desai remarked that the unit of analysis matters, with portfolio-level modeling behaving differently than analyses focused on individual facilities. Dillinger added that modernization should not be modeled solely through service-life assumptions; it represents a distinct programmatic need requiring separate analytical treatment.

ADVANCING ANALYTICAL MODELS FOR RENEWAL ESTIMATION

Workshop participants discussed the analytical models used to estimate facility renewal requirements, including both depreciation-based and condition-based approaches. They emphasized that model outputs are highly sensitive to underlying assumptions and that improving the consistency and empirical basis of these inputs is essential to producing useful estimates.

Jay Janke, DoD (retired), traced DoD’s adoption of the Facilities Sustainment Model in the early 2000s, explaining that it forecasts sustainment and recapitalization costs. He recalled that deferred-maintenance backlogs were difficult to sell to leadership because they focused on past underfunding rather than future requirements, prompting a shift toward forward-looking models based on the real property inventory. According to Janke, transparent modeling required combining the separate, incompatible real property inventories maintained by the military departments into a single, accurate DoD-wide inventory.

Lufkin, outlined an economic-depreciation approach to estimating renewal costs. He said the model uses measurable depreciation in commercial buildings as a proxy for restoration and modernization requirements. He said that, for commercial real estate, a geometric depreciation curve

appeared to provide the best fit; in that model, each year's depreciation, or loss of value, is a constant percentage of the remaining value. He compared geometric and straight-line depreciation curves and showed that they yield different reinvestment estimates. Lufkin contrasted geometric and linear models using a simple example: a 32-year service life for a standard office building produces a residual value of zero in a linear model but roughly 40 percent in a geometric one, implying that the building still retains utility and does not need total replacement. He emphasized that depreciation rates and service lives are key variables in renewal modeling and should be refined through consistent, updated data. He noted that renewal and restoration costs typically represent about 6 percent of building replacement value over time. Lufkin explained that assuming a smooth, uninterrupted depreciation curve overlooks the periodic reinvestments that extend a facility's service life, which can underestimate depreciation and skew long-term renewal requirements. He also remarked that renewal-cost estimates support federal analyses, including budgets, benefit-cost analysis for projects, and benchmark cost references.

Schmerer described how the laboratory is testing renewal cost models using 10-year projection scenarios from BUILDER SMS. By combining component-level inventory and cost data from BUILDER with site-specific analyses, LLNL is modeling deferred maintenance and modernization requirements over the coming decade. Schmerer also referenced the 2003 Facilities and Infrastructure Recapitalization Program (FIRP) audit, which independently verified deferred maintenance estimates and demonstrated the value of external model validation. He noted that after the FIRP era, NNSA implemented the BUILDER SMS in 2018, marking a shift in how deferred maintenance was defined. Through his presentation, he described depreciation-based estimating as a way to produce a defensible figure for budgeting, and he noted that varying model assumptions and scenarios can support prioritization and align plans with constrained resources.

Bartels described how DoD's SMS captures facility condition and inventory data across more than 2.2 billion square feet of assets. He noted that maintaining consistent data standards and quality control is essential to ensure that results are credible, and he added that army-wide quality assurance procedures—including standardized rules for grouping building components into assessment units, required photo

documentation, and multi-level review—are integral to maintaining data integrity. The Enterprise Sustainment Management System (ESMS), the successor to BUILDER SMS, consolidates these records and enables scenario analysis to forecast and prioritize renewal requirements. Bartels also discussed the FIA tool, which combines data from BUILDER and the Installation Status Report to calculate unconstrained requirements for restoring facilities to full condition.

Grussing explained that ERDC's SMS Technical Center of Expertise develops and manages the condition assessment tools used across DoD to collect and analyze facility data for portfolio-wide decision making. Bartels noted that these tools collect component-level data across a wide range of facility and infrastructure systems—including buildings, utilities, and waterfront and airfield structures—to support sustainment, restoration, and modernization planning. Grussing later said that future research opportunities include integrating condition data and performance indicators with cost estimation models to improve the reliability of renewal forecasts; the goal is to connect data from the SMS to other analytic tools, enabling a more comprehensive view of facility renewal and long-term planning. He pointed out that the traditional SMS focuses on condition assessment to identify sustainment needs. Renewal, however, addresses facility needs beyond just condition.

Several speakers remarked that scenario analysis methods allow agencies to simulate portfolio outcomes under different renewal strategies. Speakers highlighted testing sensitivity of backlog growth to changes in service-life parameters. Desai suggested focusing on scenario distributions rather than point predictions, while Lufkin stressed the importance of aligning on key assumptions—especially service life—and other workshop participants emphasized credibility through V&V. Several speakers reiterated that models are only as valuable as their credibility and validation, noting that the goal is usefulness rather than perfection.

Dan Sichel, Wellesley College, offered an economist's perspective on how depreciation research relates to broader questions of productivity and technological change. Using a historical example of the falling price of nails, he illustrated how advances in technology can rapidly alter the relationship among asset age, replacement cost, and economic value. He

noted that such changes can shorten effective service life and weaken the relationship between age and economic value.

Workshop participants described a shift from simpler estimating approaches to models that represent how facilities are expected to degrade over time and can express uncertainty through alternative scenarios in lieu of a single point estimate. Several speakers emphasized that depreciation-based models can provide a rigorous basis for renewal estimation, but only if service-life and depreciation assumptions are calibrated with observed condition data and validated against actual portfolio results.

INTEGRATING LARGE-SCALE DATA AND COMPUTATIONAL INFRASTRUCTURE

Workshop participants emphasized that large-scale data systems and computational infrastructure are important to scaling renewal models across large portfolios. Speakers noted that advances in analytical methods are only part of the challenge; organizations also need data systems, computing environments, and interoperable tools capable of supporting implementation, integration, and sustained use in practice.

Grussing moderated the discussion on emerging data systems and analytical tools for facility renewal modeling. He framed the session around a central question: How can existing data repositories and computational advances be used to make renewal cost estimation more accurate and more efficient?

Bartels explained that DoD's BUILDER SMS data document the age, condition, and maintenance history of federal facilities. He noted that ESMS is expanding to include utilities and other non-building infrastructure systems, ensuring consistent data structures across asset types. Bartels said that the growing volume of data presents both opportunities and challenges and that consistent data standards and rigorous quality checks are essential to maintain analytic reliability.

Deniz Besiktepe, Purdue University, discussed opportunities to integrate building information modeling, digital twin environments, and sustainment management data. She noted that although digital twin technology can provide real-time performance monitoring, its adoption in the public sector has been limited by cost, cybersecurity restrictions, and workforce readiness. She observed that facility managers often lack the personnel and time to keep digital twin data

current, noting that limited workforce capacity and complex data requirements hinder effective implementation in the public sector. She described the cultural resistance of organizations, remarking that some staff openly resist new technologies or data-sharing practices; for example, one technician told her, "I'm going to retire in five years. I really don't want to learn any more technology." Besiktepe described cases in which federal or university facility managers lacked the expertise to maintain these complex models, leading to abandoned or static data sets. She recommended starting with pilot projects, such as testing new tools on a single building or a small group of equipment, to show their effectiveness before scaling up, emphasizing that limited, well-defined efforts are more feasible for organizations that are beginning to explore technologies like digital twins.

Mark Arnone, Amazon Web Services, outlined how cloud-based infrastructure could help agencies manage and analyze large volumes of facility data. He explained that modern cloud platforms can host multi-petabyte databases and provide scalable computing resources for simulations and visualization without requiring agencies to build or maintain local data centers. He noted that cloud platforms can scale computing resources on demand, allowing agencies to pay for only the computing capacity they actually use, thus reducing the cost of large-scale simulations. Arnone noted that accredited isolated cloud environments already support sensitive federal workloads, which demonstrates their feasibility for large-scale facility data.

Besiktepe and Arnone emphasized that technology adoption must be paired with organizational readiness. Besiktepe noted that different professional groups—such as architects, engineers, and construction and facilities staff—often work in isolation, making it difficult for organizations to integrate new technologies effectively. Arnone emphasized that strong data governance, including validation of data accuracy and consistency, is essential for agencies to use cloud technologies effectively, regardless of the specific hardware or software involved.

Workshop planning committee member Michael Eriksen, Purdue University, who moderated the session on translating theory to practice, invited representatives from JLL Technologies (JLLT) to discuss how large organizations are applying predictive analytics and real-time data to guide

facility investment decisions. Michael Ewert, JLLT, explained that commercial portfolio managers increasingly rely on depreciation-based forecasting to prioritize investments and support funding requests. Other JLLT panelists added that such forecasting depends on continuously updated condition data and time-based monitoring to keep analyses current. They emphasized that maintaining accurate and validated information is more important than simply accumulating large volumes of data. Ewert and other JLLT panelists described how analytical frameworks that combine renewal cost modeling with real-time asset data allow commercial managers to align forecasting with day-to-day portfolio decisions. Martine Combal, JLLT, emphasized the importance of aligning facility data with financial and leasing information so that renewal decisions reflect both physical condition and business objectives. She observed that agencies frequently underestimate the effort needed to harmonize disparate data sources and noted that fragmented information across systems can make portfolio planning and investment analysis difficult. Combal noted that linking leasing and facility data allows managers to identify under-used space and reduce portfolio size proactively. Yann Palmore, JLLT, described emerging practices that merge equipment-level telemetry with inspection data to create continuously updated performance models. Palmore explained that telemetry allows operators to understand how equipment performs under real-world conditions, improving maintenance planning and resource allocation. He outlined real-time models that automatically update as sensors capture data on equipment use, maintenance activity, and environmental conditions, providing a more accurate picture of asset condition. Peter Scanlon and Jim Barlow, JLLT, added that continuous data exchange among design, construction, and operations platforms can eliminate information loss at project handoffs. They noted that such integration enables organizations to move from static, point-in-time condition assessments to continuously updated models of asset performance that also draw on historical data, providing a more accurate basis for maintenance and investment planning. These sorts of systems, they pointed out, would both reduce the need for manual data collection and evaluate future scenarios more efficiently.

Eriksen summarized that these private-sector experiences illustrate how analytical and technological advances can

close the gap between theoretical modeling and operational decision making.

The discussion highlighted how analytic tools now used in large commercial portfolios—real-time monitoring systems, automated depreciation updates, and continuous data exchange—mirror capabilities that federal agencies are beginning to test within their own SMSs. Workshop participants noted that these advances make it increasingly feasible to apply similar computational techniques to public-sector portfolios once data standards and governance structures are in place. Several speakers identified the lack of interoperability among asset management, financial, and design systems as a persistent obstacle. Condition data often reside in BUILDER or other sustainment platforms, while financial data are stored separately in budgeting systems and design data remain isolated within project or modeling software. Linking these sources requires standardized identifiers and shared taxonomies for building systems and components. Grussing emphasized the importance of leveraging existing federal data sources to support facility renewal planning and cost estimation.

In closing, speakers highlighted that the technical challenges of integrating large-scale facility data are increasingly solvable, while the institutional and governance challenges remain substantial. They emphasized that the analytical payoff of integration, including improved model calibration and credible uncertainty analysis through verification and scenario testing, will depend on sustained collaboration across the federal enterprise. Besiktepe warned that ongoing implementation will depend on workforce capacity, including training and having personnel to keep integrated data systems current over time.

ESTABLISHING CREDIBLE AND VERIFIABLE MODELS

Gowri Srinivasan, Los Alamos National Laboratory (LANL), explained that developing confidence in analytic models requires systematic V&V. Drawing on two decades of experience verifying high-consequence physics codes at NNSA's laboratories, she said that V&V is “the process of demonstrating credibility.” The goal, she noted, is not to save money or shorten analysis time but to establish confidence in model results and the decisions informed by them.

Srinivasan described verification as checking that the model was implemented as intended—confirming that equations, algorithms, and logic are coded correctly—and validation as determining how well the model represents reality for its intended use. She emphasized that distinction between verification and validation matters because even a well-constructed model can fail when used outside its designed range. She added that users and developers share responsibility for ensuring that a model is applied only within the conditions for which it was designed. She explained that high-consequence applications warrant independent peer review to verify credibility and that validation should include comparison with experimental or field data whenever available.

To illustrate how intended use and consequence affect V&V, Srinivasan explained that NNSA laboratories perform both low- and high-consequence modeling. For exploratory or preliminary studies, she said, analysts often apply “good-enough” checks—that is, visual or surface-level reviews used to spot obvious errors or implausible results. These quick screenings can narrow hundreds or thousands of possibilities to a few promising cases, which then undergo deeper V&V to ensure the models produce the right answers for the right reasons. She noted that the level of V&V should scale with the consequences of the application so that the rigor of review matches the importance of the decisions the model will inform.

She offered several practical recommendations drawn from national laboratory experience. Modelers should test both routine and corner-case conditions to identify where logic fails, using what-if scenarios to expose black-swan or rare-event risks. Validation data, she said, should be refreshed as new information becomes available, and assumptions, parameter ranges, and uncertainties should be documented in sufficient detail. She cautioned that a model’s logic may hold only within its validated range and warned that results can break down when applied beyond the region for which the model was tested. At LANL, she added, these practices are embedded in a culture of continual testing and “postdiction”—in other words, comparing previous forecasts to actual outcomes to evaluate how well a model performed.

Srinivasan stressed that these standards are routine for safety-critical nuclear simulations, where experiments

cannot always be performed, and that rigorous V&V is essential for establishing credibility in such models. V&V, she said, informs decision makers of the confidence and risk associated with a model’s results. Lufkin added that the renewal cost model should be formally verified and validated to confirm that it performs as intended and accurately reflects observed conditions before agencies apply it broadly.

Srinivasan concluded that V&V is never finished; it is a continuing obligation that evolves with new information. She said that every model run should be viewed as an experiment whose results can improve subsequent forecasts, turning model use itself into a process of learning and refinement.

FUTURE RESEARCH AND DEVELOPMENT PRIORITIES

Across multiple sessions, workshop participants identified technical R&D priorities that could improve the accuracy, transparency, and usefulness of renewal cost models. They emphasized the merit of improving the quality and consistency of existing modeling approaches and data resources.

Lufkin reiterated that much of the empirical basis for current depreciation and service-life assumptions comes from studies conducted decades ago. He encouraged updating the empirical data on depreciation rates and service lives to improve the accuracy of renewal estimates. Lufkin advised that agencies agree on common service-life definitions before collecting new data to avoid incompatible baselines. He observed that updating these parameters would benefit every major modeling approach—geometric, linear, or condition-based—because each depends on the same underlying inputs.

Srinivasan suggested that model developers incorporate formal uncertainty quantification techniques that are common in computational science. She emphasized that V&V should be treated as a continuous process, with new data and test cases added over time to strengthen model credibility. Schmerer described how integrated modeling approaches that link condition data, depreciation rates, and investment scenarios can help visualize the effects of different funding levels and service-life assumptions. Desai emphasized that future research should demonstrate due diligence by clearly defining data sources, assumptions, and model structures. He noted that agreement on these foundational elements is essential for credible results and their use.

Dillinger pointed out that climate-driven environmental changes, such as permafrost melt, will require updated design standards and modeling assumptions for infrastructure renewal. Speakers identified emerging technical opportunities, including the use of machine learning to analyze inspection data, the application of real-time sensor information to improve condition assessments, and the development of broader “enterprise models” that connect facility planning with mission and workforce considerations. Workshop participants noted that recent advances in artificial intelligence allow analysts to generate prioritized renewal options within minutes, dramatically accelerating scenario development and decision support. Taken together, these research directions would move renewal modeling from static estimation toward a more adaptive analytical discipline, in which parameters evolve as empirical evidence accumulates and verification is continuous rather than periodic.

MODEL APPLICATION AND DECISION SUPPORT

Finally, speakers throughout the workshop discussed how technically sophisticated models can inform real-world investment decisions without overwhelming decision makers. The challenge, participants noted, is not only producing accurate forecasts and communicating their implications clearly but also presenting renewal needs in a form that fits the constraints and structures of the federal budget process.

David Haun, former OMB, explained that large-scale renewal decisions ultimately depend on how long-term capital costs are captured in the federal budget process. He noted that although total discretionary spending exceeds \$1 trillion annually, only a small fraction—about 2 percent—is directed to capital reinvestment. Haun pointed out that this limited share reflects not an absence of resources but rather choices about allocation. Emphasizing that agencies must account for the full long-term costs of ownership and maintenance when making investment decisions, he cited past efforts to embed such costs in federal budgeting structures, including requirements for agencies to contribute to employee retirement funds and proposals for a federal capital revolving fund.

Skip Vaughn, American University and former head of facilities at the National Institute of Standards and Technology (NIST), noted that renewal programs often emphasize keeping facilities operational for as long as possible with

limited resources rather than addressing broader performance limitations. He also described how NIST lacked formal definitions for classifying work and how congressional appropriations categories often drove project categorization, shaping what could be funded as deferred maintenance versus capital construction.

Crutchfield explained that decision makers must weigh renewal investments against competing priorities, often within a single fiscal year. Dempsey described how agencies facing chronic resource constraints often concentrate on their highest-priority needs, a “management of scarcity” doctrine that he said can be effective under these restrictions. Dempsey noted that agencies often face more statutory and mission requirements than available funding can meet, forcing difficult prioritization decisions.

Cash Fitzpatrick, NNSA, discussed how analytical results must be presented clearly and concisely for decision makers and cautioned that overly complex metrics or excessive detail can obscure the message. He emphasized that while data, modeling, and metrics are necessary to establish credibility, they are not sufficient to make a compelling budget case without clear communication. Fitzpatrick advised analysts to focus on a few well-chosen examples that demonstrate how data are used to improve decisions and to prepare for close questioning about their methods and results. Crutchfield added that effective graphics can strengthen the message, helping leaders grasp key points quickly and see the connection between data and mission outcomes.

Desai explained that differences among models can arise from both assumptions and data and that clarifying these sources of variation is key to understanding uncertainty. He noted that modelers should explain not only their data and assumptions but also the level of aggregation used—whether the model represents individual facilities or an entire portfolio—because that choice affects how results are interpreted.

Conger highlighted that extending a facility’s assumed service life can dramatically alter modeled outcomes for deferred maintenance. He cautioned that if such results are shown without context, decision makers might misinterpret them to mean that assets can simply be used longer to avoid costs. Conger emphasized that models should make clear the

real-world implications of service-life assumptions and the consequences of allowing systems to fail.

Grussing observed that assessing a facility's functionality is often more difficult than assessing its physical condition because functionality depends on how well a building supports its mission, rather than simply on whether it is in good repair. He noted that evaluating functionality requires input from mission operators as well as engineers, because renewal decisions depend on how facilities are used to accomplish mission goals. Grussing added that considering functionality alongside condition helps agencies connect physical renewal decisions to mission performance, and he emphasized the importance of collaboration among engineers, operators, and planners to align assessments with mission needs.

Several speakers identified opportunities to improve consistency across existing renewal cost models. They pointed to the use of similar assumptions for service life and depreciation rates, better alignment of data definitions and structures, and more compatible reporting formats as practical ways to enhance comparability across agencies. They emphasized that the goal is not to create a single monolithic tool but rather to develop compatible methods and data practices that can evolve incrementally while improving the transparency and comparability of renewal estimates across agencies.

In total, these discussions underscored that improving renewal modeling will depend not only on refining technical approaches, but also on strengthening the data, assumptions, and analytical frameworks needed to produce results that inform decision making.

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